HARVARD MEDICAL SCHOOL | BLAVATNIK INSTITUTE BIOMEDICAL INFORMATICS

HARVARD School of Dental Medicine

Artificial Intelligence in Dental Medicine Augmented Diagnostics and Computer Vision

Part 1 06/24/2026

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whoami

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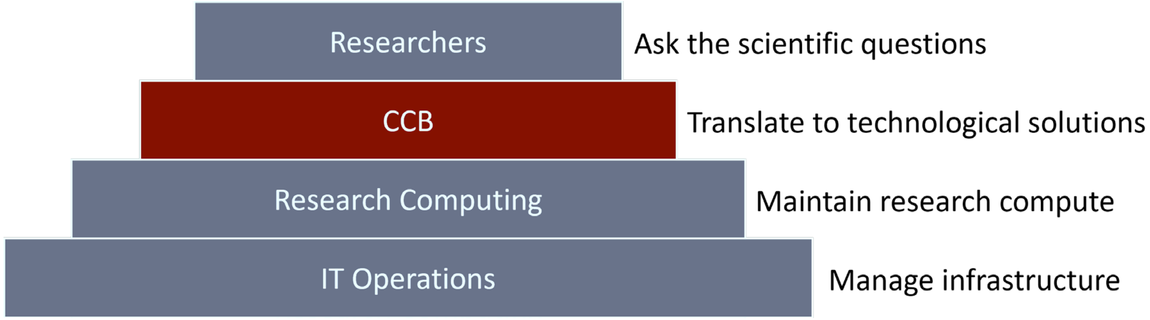
- **Analytic support and consulting**
- **Research, education, and administration**
- **Collaborate on research projects**
- **Help with access to computational resources**



Agenda

Part 1: Introduction, data, and testing

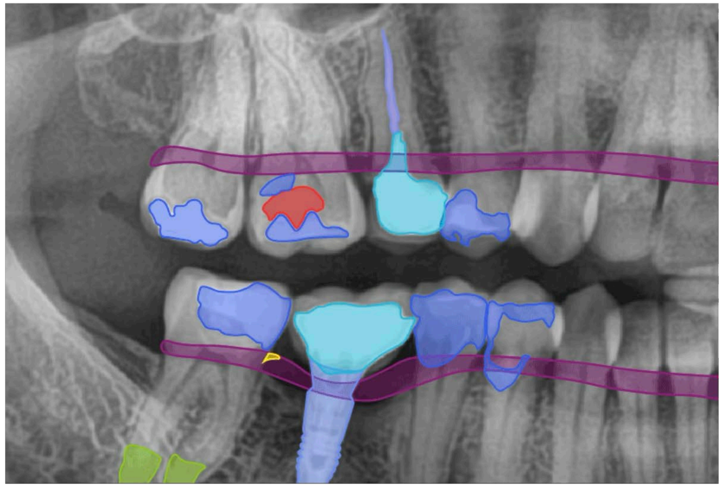
- Computer vision and AI
- Getting data for training models
- Working with clinical data, PHI
- Part 2: Training a vision transformer model
- Evaluating model performance

Part 2: Training a detection transformer model

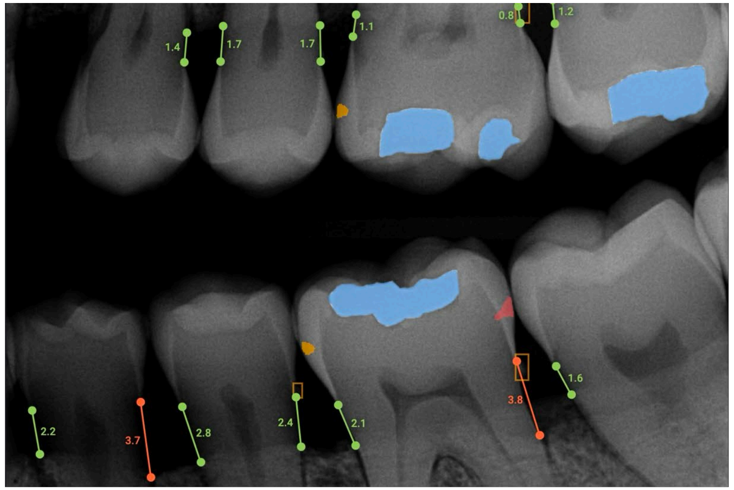
- More “technical” session than part 1
- Goal: provide resources to train your own model
- Getting started with code

  The computer vision repository
- DETR: State-of-the-art detection transformer
- Model training process
- Use your own data

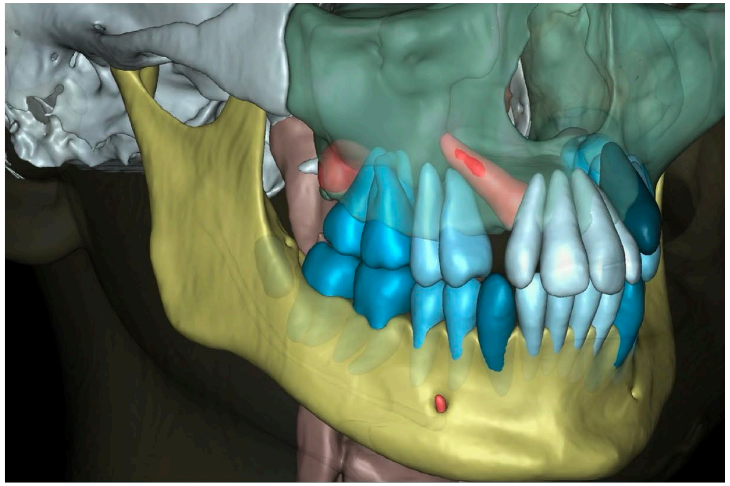
Dental radiography in the US



Panoramic radiography
dentalXr.ai



Intraoral radiography
Overjet



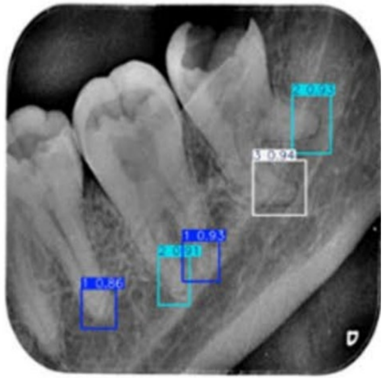
Cone beam computed tomography
Diagnocat

- Produces one out of every four diagnostic radiographs
- > 300 million dental images per year in the US
- ~ 30% of the world's total dental radiography procedures

Patient exposure from radiologic and nuclear medicine procedures in the United States and Worldwide
Mahesh et al. Radiology (2023)

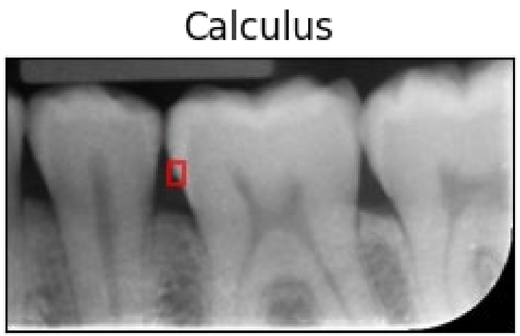
Large datasets are generated every year and could be used to train capable expert models

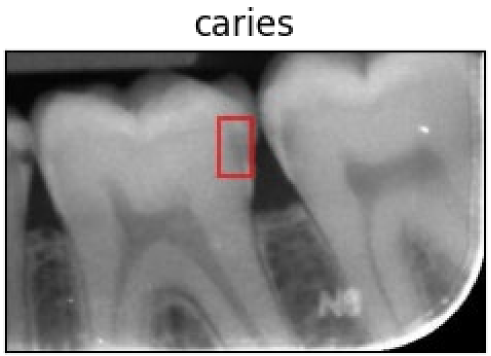
Primary tasks for computer vision

Task	Use case		Model output
Image classification	“Is the quality good enough, or do I need to retake it?”		Class label: “Incomplete apical coverage” Deep learning-based assessment of periapical radiographic image quality. Chi et al. Nature Scientific Reports (2026)
Object detection	“Find and classify apical periodontitis”		Bounding boxes with periapical index (PAI) scores Automated assessment of periapical health based on the radiographic index using one-stage object detection algorithms. Saber et al. Nature Scientific Reports (2025)
Segmentation	“Measure periodontal bone loss”		Radiographic bone level (RBL) percentage rate Automatic recognition of teeth and periodontal bone loss measurement in digital radiographs using deep-learning artificial intelligence. Chen et al. J. Dent. Sc. (2023)

Computer vision in dental practice

Benefits for patients and providers





- Help detect subtle findings
- Provide reproducible measurements
  (bone levels, root lengths)
- Increase objectivity / reduced human bias (applying the same criteria to every image)
- Allows effective screening and efficient clinical workflows
- Reduce uncomfortable/invasive procedures
- Better patient communication

How does computer vision fit into the context of AI?

Computer vision

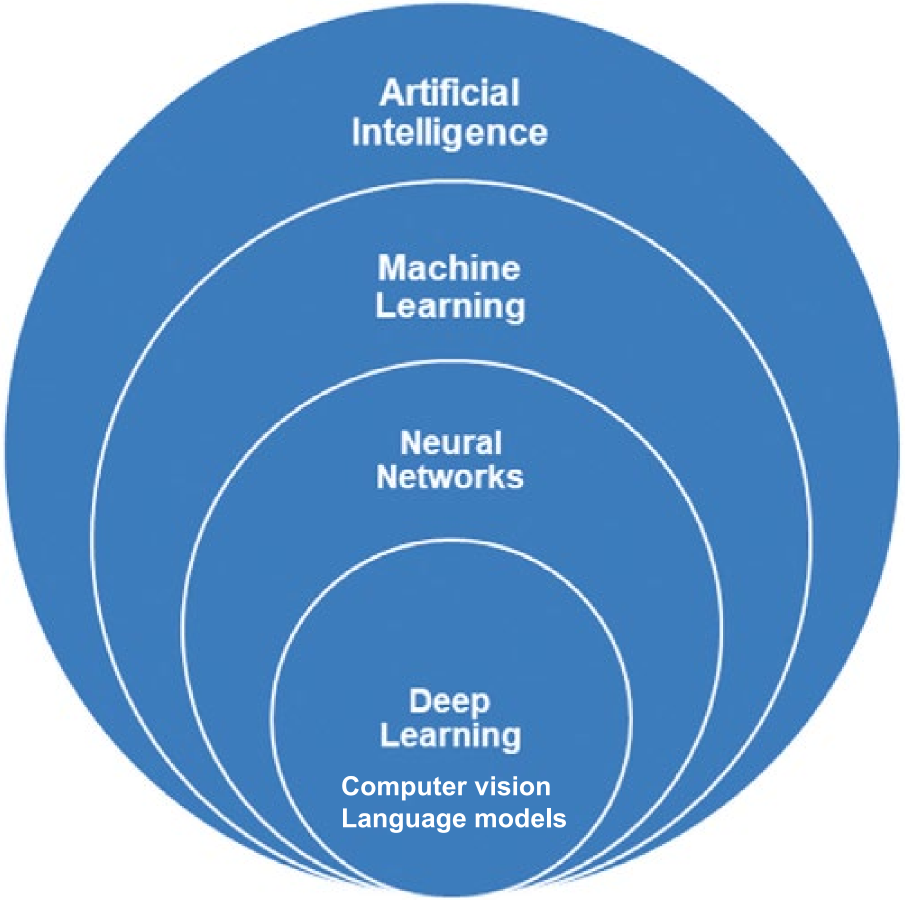
How does it fit into the context of AI?

AI Overview

artificial intelligence

/ˌärdəˌfiSH(ə)l ənˈteləj(ə)n(t)s/

Artificial intelligence (AI) is a branch of computer science focused on building systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, and recognizing patterns. Merriam-Webster +2



Applications, functions and accuracy of artificial intelligence in restorative dentistry
Tabatabaian et al. J Esthet Restor Dent. (2023)

Includes many types of algorithms

- Classical machine learning (regression, decision trees, …)
- Neural Networks
- Deep neural networks

  Image models (image analysis, generation)

  Large Language Models (ChatGPT, Claude, Gemeni)

Neural networks for image data

Why we cannot use a linear regression model for computer vision

Classical ML does not work very well

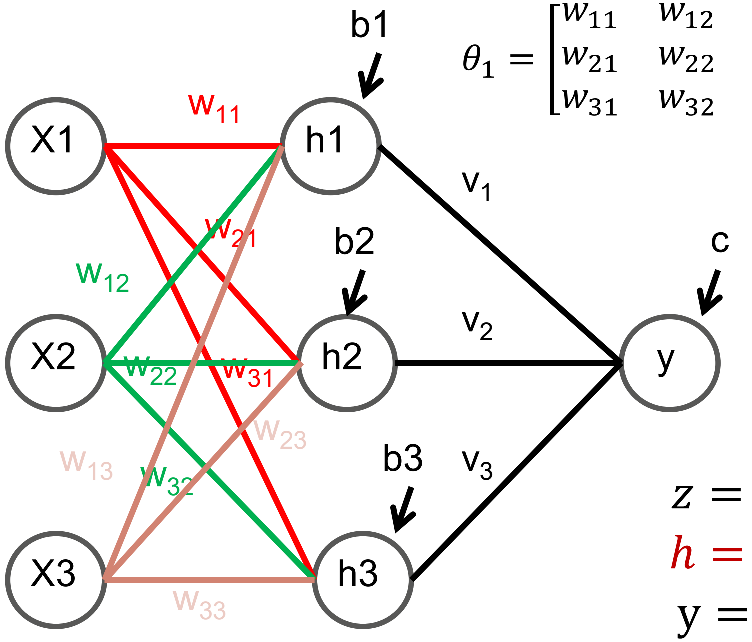
- Pixel intensities - shapes are highly non-linear (caries, fillings, crowns)
- Pattern recognition requires combinations of neighboring pixels
- Linear regression would treat each pixel independently
- Cannot capture complex shapes

Artificial neural networks

- Special type of machine learning algorithm
- Described in the early 1960s by Frank Rosenblatt
- Computing model inspired by the human brain
- Consists of layers of interconnected nodes (neurons)
- Nodes that can reinforce or suppress signals
- Training adjusts the weight of each connection
- Learns non-linear relationships in data

Artificial neural network

$$\theta_1 = \begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix} \quad \text{b} = \begin{pmatrix} b1 \\ b2 \\ b3 \end{pmatrix}$$



$$z = \boldsymbol{x} * \theta_1 + b$$
$$h = sigm(z)$$
$$y = \boldsymbol{v} * \boldsymbol{h} + c$$

Input layer
Pixel intensities

Hidden layer
Patterns

Output layer
Diagnosis

https://en.wikipedia.org/wiki/Perceptron

Neural network architectures

Convolutional Neural Network CNN

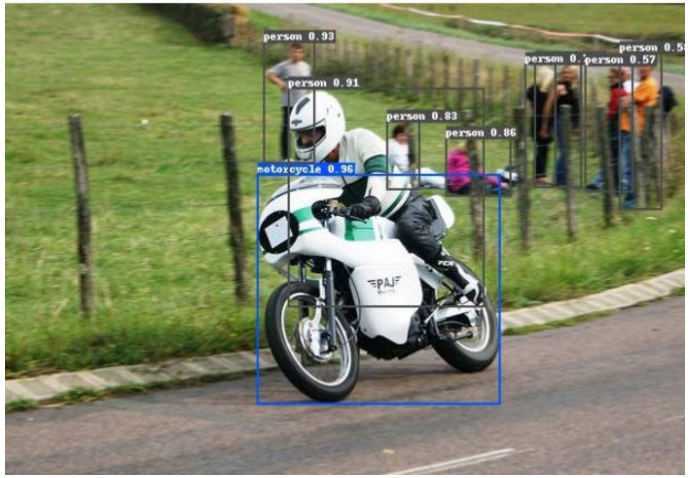
- **1998 Yann LeCun ANN LeNet-5 model (60K parameters)**
- 2000 Traditional ML (high computational cost of ANNs)
- **2012 Deep learning revolution: AlexNet**
- 2015 Classification models (surpassed human abilities)
- 2016 Object detection: YOLO
- 2018 Segmentation and detection: Detectron
- 2020 Object detection with transformers
- 2024 Detection transformers beat YOLOs: DETRv2

Current research: fast and accurate segmentation models

CNNs continue to be important. They are integrated components of latest architectures (transformers)

MNIST dataset
LeNet 1998

Hand-written postal zip codes

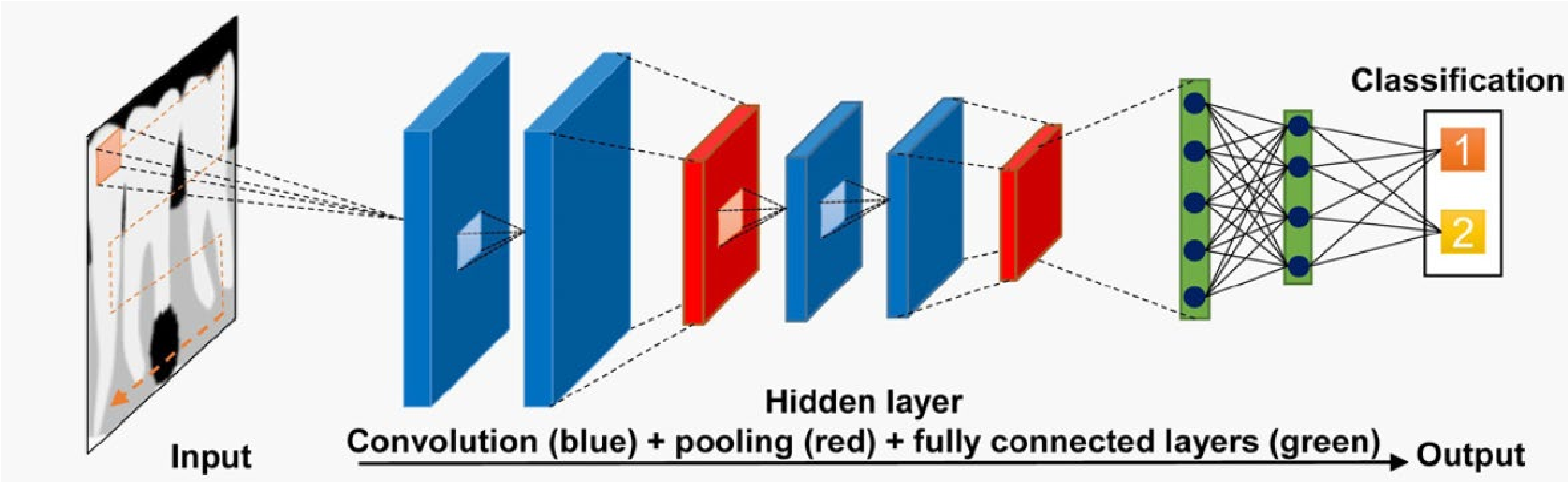


Object detection
DETRv2 2024

https://en.wikipedia.org/wiki/Neural_network_(machine_learning)

Convolutional neural networks

It's worth understanding it

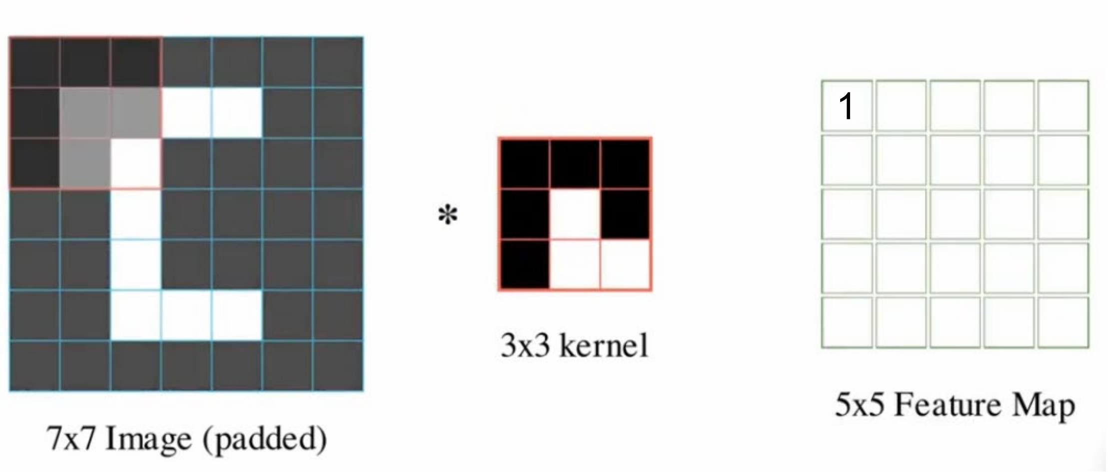


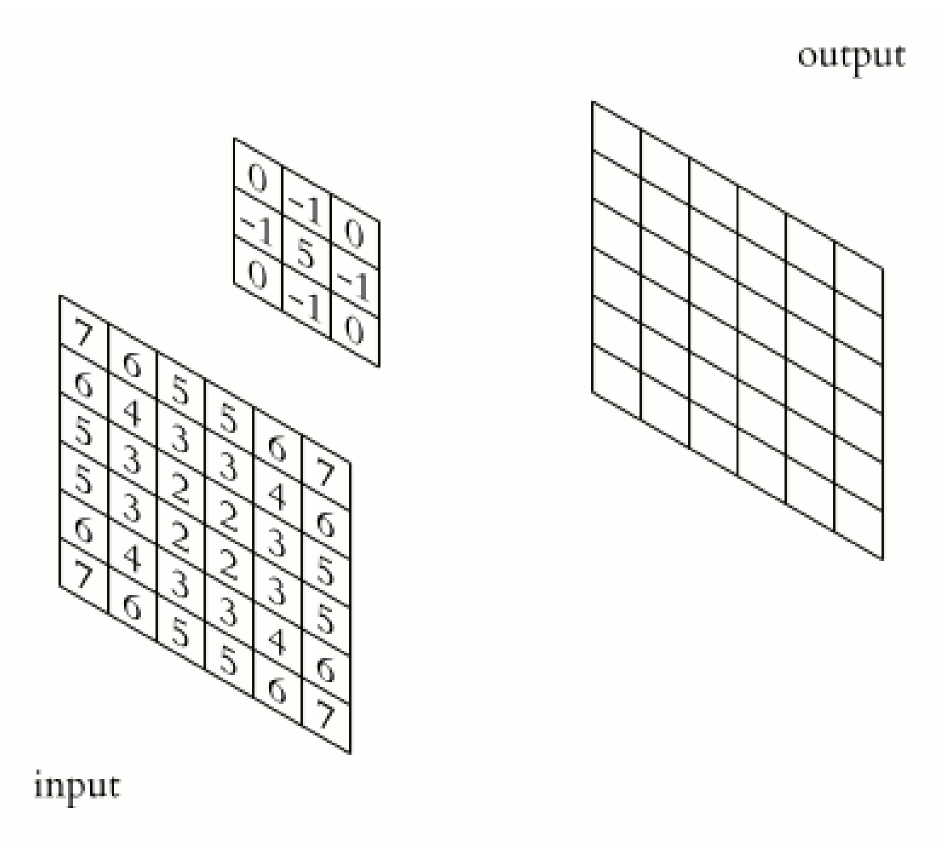
- Most successful neural network architecture for image analysis
- Designed to preserve the 2D nature of images
- Shared weights (feature map) trained to recognize patterns (edges, lines, curves)
- Processes the input image with a grid-like structure by scanning it with small filters

Putra et al. Current applications and development of artificial intelligence for digital dental radiography (2022)

Convolutional neural networks

How the model is trained on images





Convolution operation

- Scan filters across the image and calculate the product
- Amplifies signal where the filter matches the image
- Attenuates signal where there is little overlap
- Stack multiple convolutional layers -> CNN

https://medium.com/thedeephub/convolutional-neural-networks-a-comprehensive-guide-5cc0b5eae175

The theory and practice of computer vision

Some useful resources

YouTube
CS lectures from Stanford

https://youtube.com/playlist?list=PLoROMvodv4rOmsNzYBMe0gJY2XS8AQg16&si=IKSqlcl2VLroesEP

Online learning platforms
Excellent for hands-on projects

https://www.udemy.com/share/105H86/

Online tutorials
Exist, but good ones are hard to find

https://blog.roboflow.com/train-rt-detr-custom-dataset-transformers/

Let’s talk about data

How much data do I need?

- Model architecture
- Size of the model
- Difficulty of the task

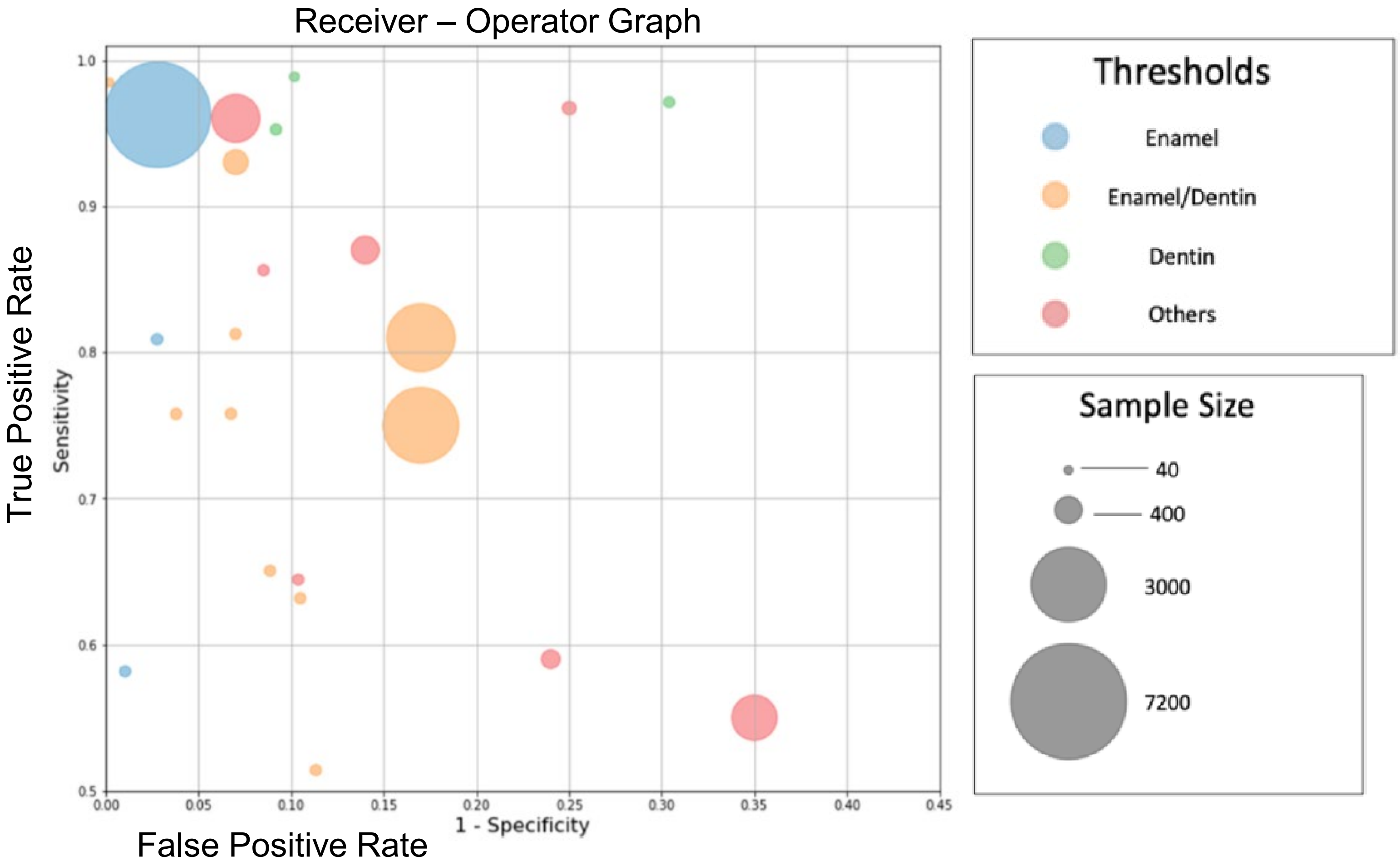
    Restorations

    Caries

    Bone levels

    Periodontal disease

How do we get data?



Mohammad-Rahimi et al: Deep learning for caries detection: A systematic review. J. Dentistry (2022)

Working with clinical data

Electronic Health Records

- Electronic version of medical history
- Personal information (name, age, sex, address)
- Medications
- Medical problems
- Exam results
- Lab results
- Progress notes

Image Data

- Document diagnosis
- Treatment results
- X-ray images (intraoral, panoramic)
- CT scans
- Photographs

The data

- Stored on secure clinical servers in various locations
- Contain personal identifiable information (name, address, birthdate, SSN, insurance, billing)
- Configured for clinical practice
- Not accessible for research
- Not suitable for large-scale data access

Working with Protected Health Information

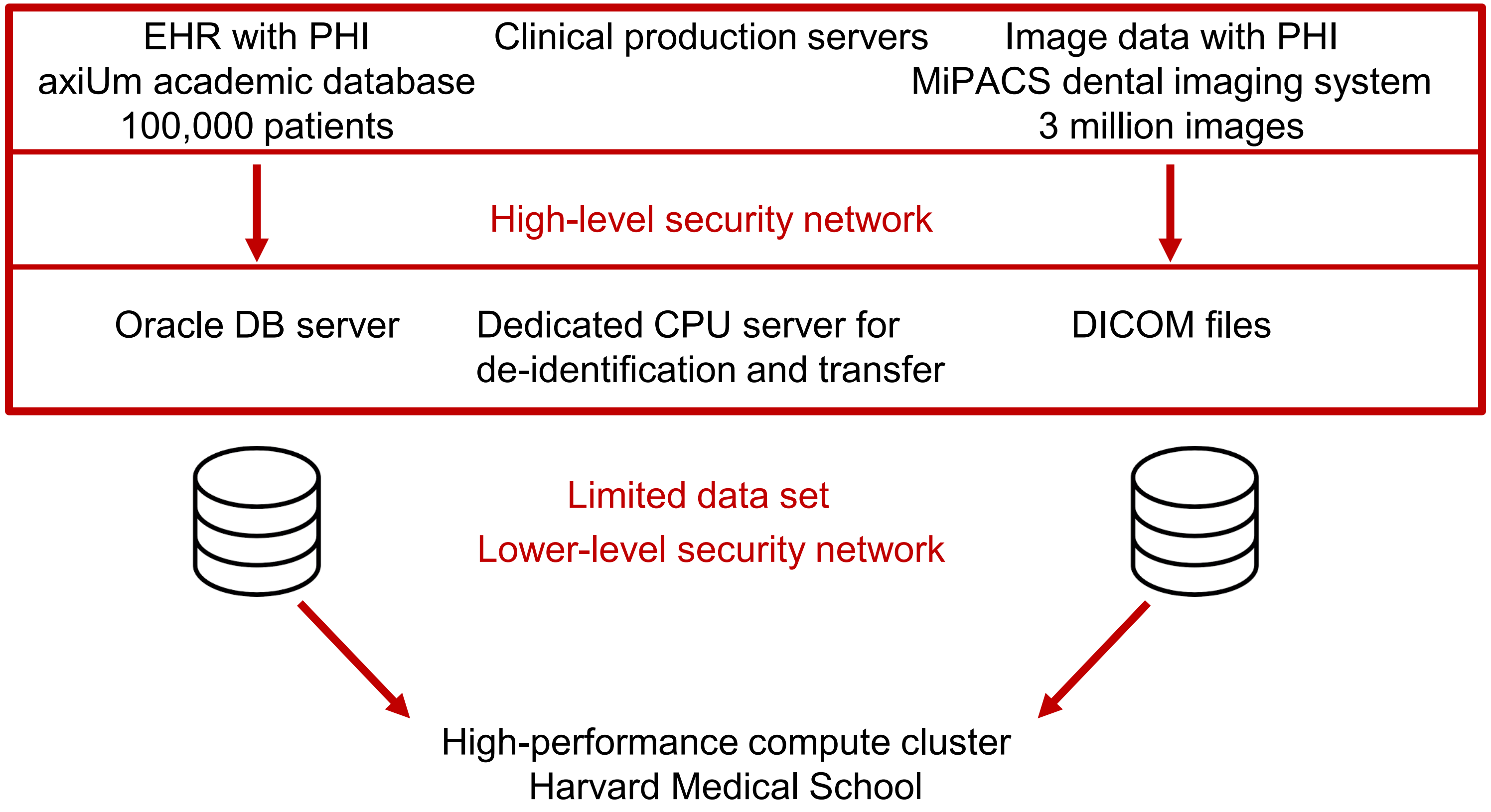
Getting the data is the most difficult step

Approach

- Copy EHR DB and image data
- Remove patient identifiers
    - limited data set (if allowed)
    - de-identified data set
- Copy de-identified data to compute
    - Images, data frames, database
    - On-premises, cloud
- Access to GPU hardware

Dealing with PHI

- Work with the IRB office
- Contact data security and IT compliance



https://www.hipaajournal.com/limited-data-set-under-hipaa/
https://www.hhs.gov/hipaa/index.html

Model workflow

Creating a computer vision model from scratch

- Research idea
- Identify the patient cohort
- If it involves Protected Health Information (PHI)
  - IRB approval
  - Secure location to work with PHI
  - De-identification
- Locate a compute environment with GPU access
- Prepare data sets for training and testing
- Train the model
- Improve model, hyperparameter tuning

Getting access to the data and preparing the data sets (de-identification) might take 70 % of the time for a project

You want to test an idea for a computer vision model?

And / or

You don't have access to patient data?

Sources for public data sets

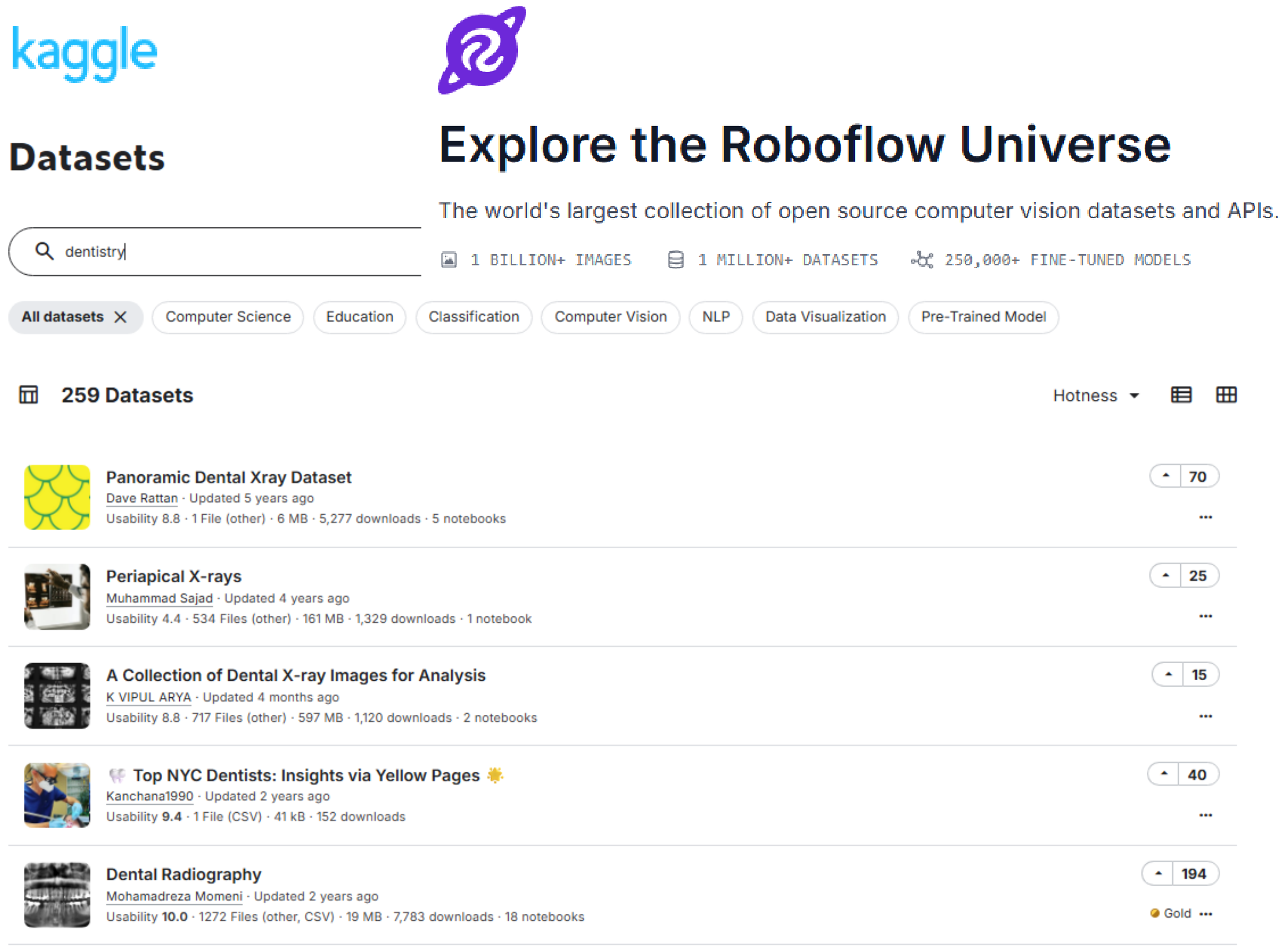
Getting started without worrying about patient privacy

Good places to find data:

- Kaggle: https://www.kaggle.com/
- Zenodo: https://zenodo.org/
- Roboflow: https://universe.roboflow.com/
- Grand Challenge: https://grand-challenge.org/
- Hugging Face: https://huggingface.co/datasets

Public data sets: .jpg, png, .csv , .json

Clinical data sets: DICOM



DICOM files

Digital Imaging and Communications in Medicine

- Standard format for medical imaging
- Contains pixel data
- Many attributes
    - Imaging parameters
    - Modality (panoramic, intra-oral)
    - Procedure codes, dates
    - Patient identifiers (name, patient ID, dob, …)
- Often only place where PHI is stored

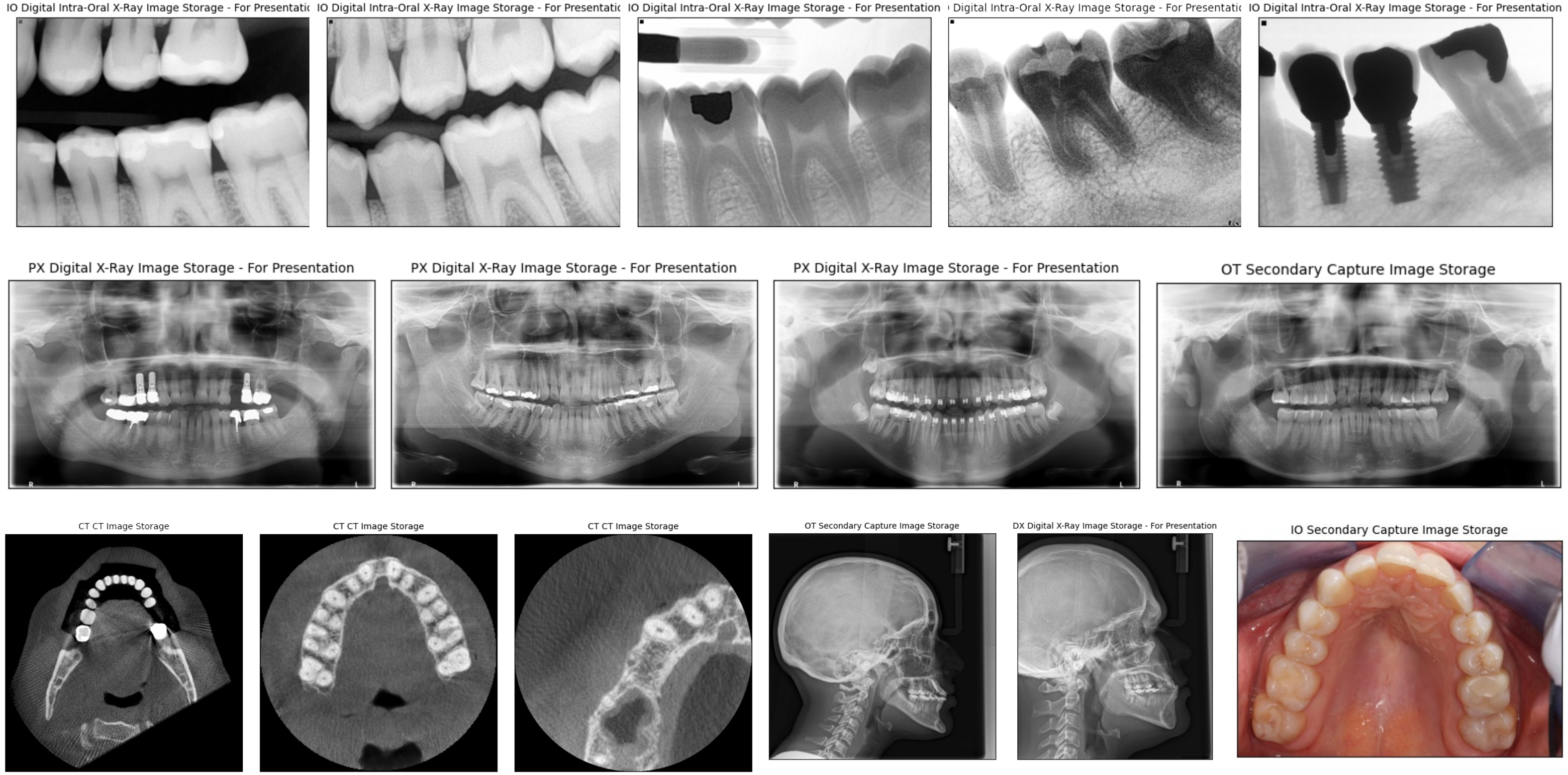
```
                   Filename: 'C:\Program Files\MATLAB\R2021b\toolbox\images\imdata\C
                FileModDate: '18-Dec-2000 12:06:43'
                   FileSize: 525436
                     Format: 'DICOM'
              FormatVersion: 3
                      Width: 512
                     Height: 512
                   BitDepth: 16
                  ColorType: 'grayscale'
FileMetaInformationGroupLength: 192
  FileMetaInformationVersion: [2×1 uint8]
     MediaStorageSOPClassUID: '1.2.840.10008.5.1.4.1.1.7'
  MediaStorageSOPInstanceUID: '1.2.840.113619.2.1.2411.1031152382.365.1.736169244'
          TransferSyntaxUID: '1.2.840.10008.1.2'
      ImplementationClassUID: '1.2.840.113619.6.5'
   ImplementationVersionName: '1_2_5'
SourceApplicationEntityTitle: 'CTN_STORAGE'
      IdentifyingGroupLength: 414
                  ImageType: 'DERIVED\SECONDARY\3D'
                SOPClassUID: '1.2.840.10008.5.1.4.1.1.7'
             SOPInstanceUID: '1.2.840.113 ... 2382.365.1.736169244'
```

PHI in DICOM data

- If ONLY in attributes (metadata):
    - Easy: just replace them
- If on the images
    - Difficult: detect and mask PHI

https://pydicom.github.io/pydicom/dev/

Extract image data from DICOM files



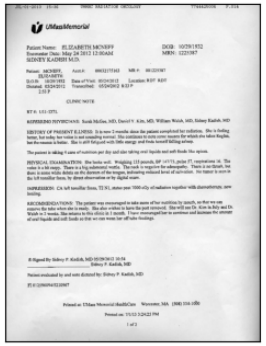
- Extract metadata and images
- Categorize by modality (IO, PX, CT, ..)
- Match patients with model cohort
- Save to standard image format (.png)
- Beware of PHI where it should not be

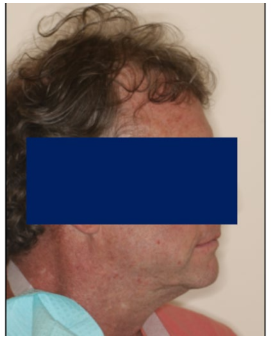
  Text on images (rare)

  Photographs

  Screenshots of provider letters

Imaging modality: IO





How do you pick the model?

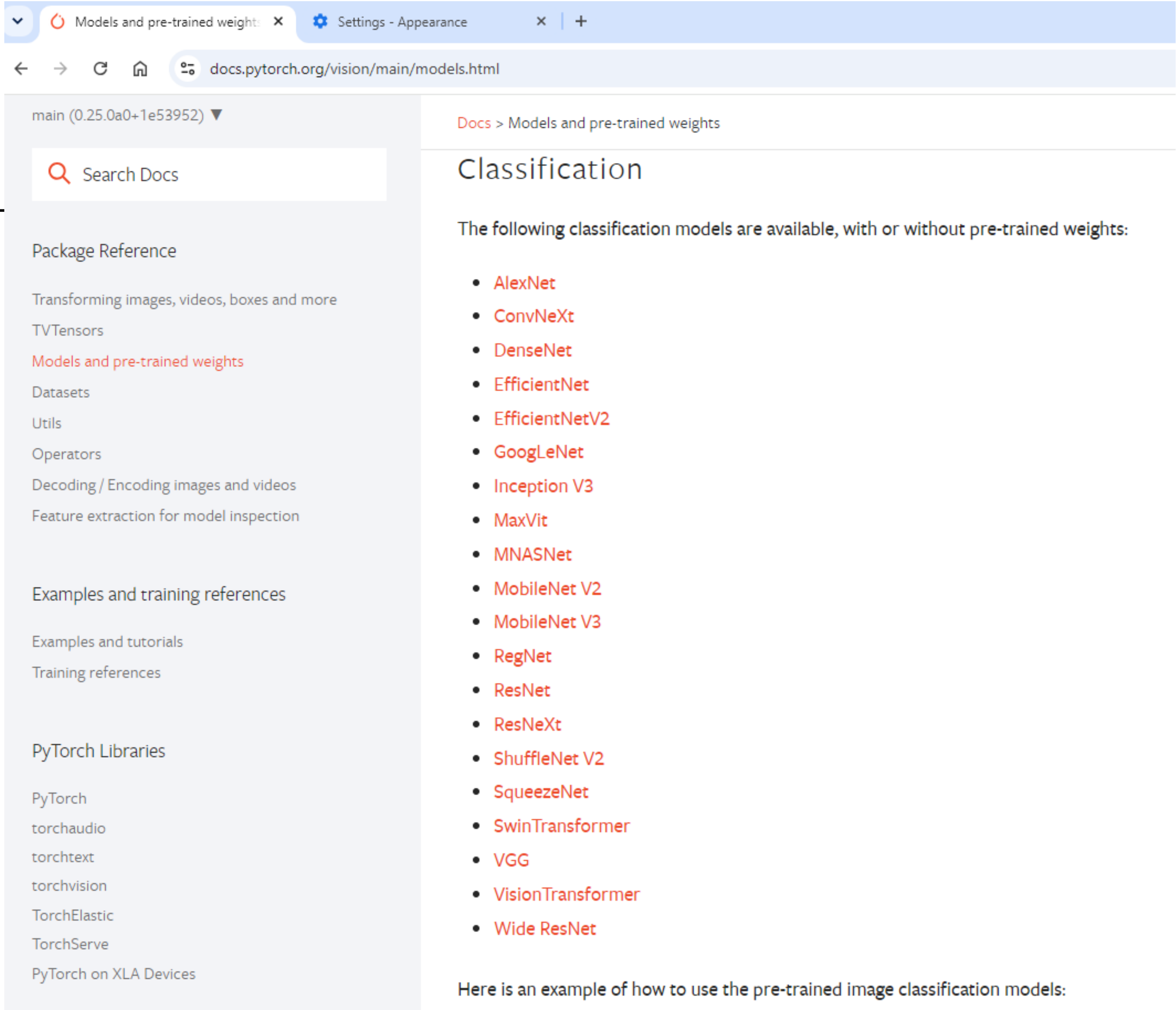
The model architecture

How do I know what model works best?

Bad news: You don't know

Good news: It doesn't matter

- If the feature you are trying to detect is in an image, any modern algorithm can be trained to detect it.
- Once you have a data set and code, the architecture is easy to switch


Models and pre-trained weights
Settings - Appearance
docs.pytorch.org/vision/main/models.html
main (0.25.0a0+1e53952) ▼
Search Docs

Package Reference

Transforming images, videos, boxes and more
TVTensors
Models and pre-trained weights
Datasets
Utils
Operators
Decoding / Encoding images and videos
Feature extraction for model inspection

Examples and training references

Examples and tutorials
Training references

PyTorch Libraries

PyTorch
torchaudio
torchtext
torchvision
TorchElastic
TorchServe
PyTorch on XLA Devices

Docs > Models and pre-trained weights

Classification

The following classification models are available, with or without pre-trained weights:

- AlexNet
- ConvNeXt
- DenseNet
- EfficientNet
- EfficientNetV2
- GoogLeNet
- Inception V3
- MaxVit
- MNASNet
- MobileNet V2
- MobileNet V3
- RegNet
- ResNet
- ResNeXt
- ShuffleNet V2
- SqueezeNet
- SwinTransformer
- VGG
- VisionTransformer
- Wide ResNet

Here is an example of how to use the pre-trained image classification models:

Training process

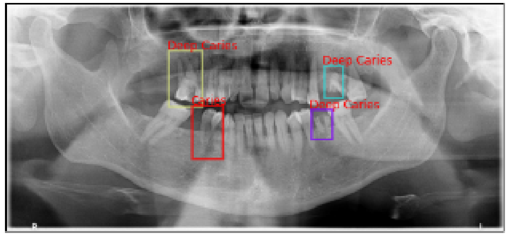
The Computer Vision Repository

The CCB Computer Vision Repository
https://github.com/ccb-hms/computervision

- Manually curated code repository
- End-to-end model implementations
- Open-source libraries and data sets
- Code templates
  - Image classification (ResNet50)
  - Instance segmentation (Mask R-CNN)
  - Object detection (RT-DETRv2)
  - Image-Language learning (coming soon)

More details in part 2!

python 3.12 | pytest passing | docker passing



The CCB Computer Vision Code Repository

This repository contains template code that that can be used as a starting point for computer vision projects. All frameworks, libraries, and data sets are open source and publicly available. Some common tasks included here are:

- Image Classification
- Object Detection
- Instance segmentation
- Gradient-weighted Class Activation Mapping

The Dentex Challenge 2023

The Dentex Challenge 2023 aims to provide insights into the effectiveness of AI in dental radiology analysis and its potential to improve dental practice by comparing frameworks that simultaneously point out abnormal teeth with dental enumeration and associated diagnosis on panoramic dental X-rays. The dataset comprises panoramic dental X-rays obtained from three different institutions using standard clinical conditions but varying equipment and imaging protocols, resulting in diverse image quality reflecting heterogeneous clinical practice. It includes X-rays from patients aged 12 and above, randomly selected from the hospital's database to ensure patient privacy and confidentiality. A detailed description of the data and the annotation protocol can be found on the Dentex Challenge website. The data set is publicly available for download from the Zenodo open-access data repository.

Getting started with this repository

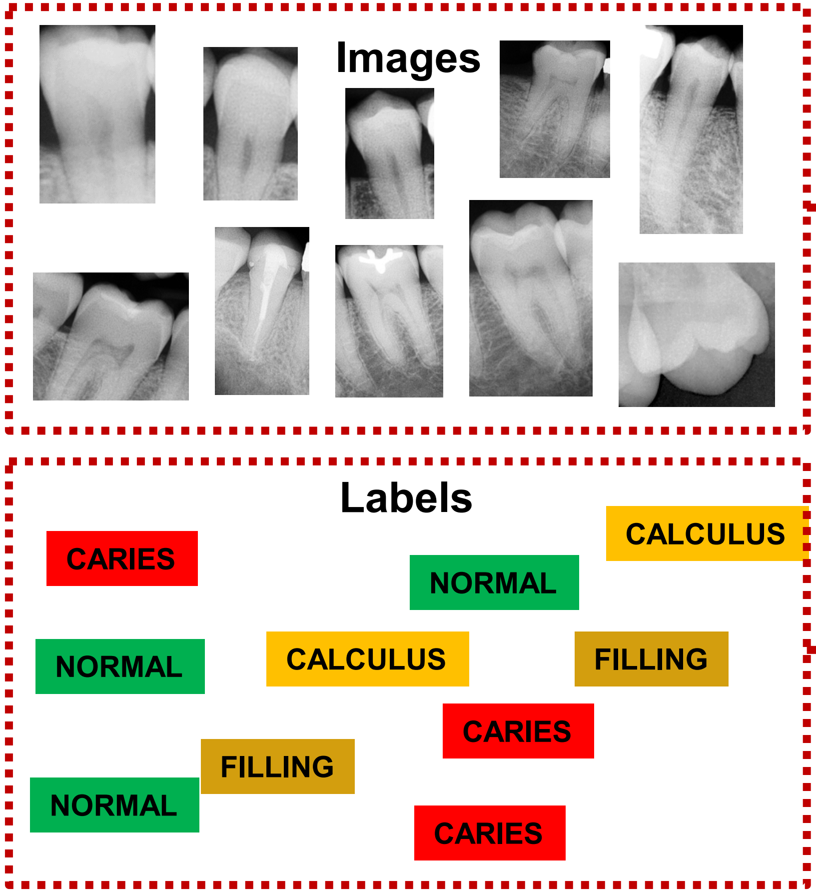
Install on your local machine

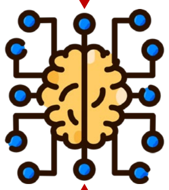
Install on the HMS O2 cluster

Label Studio

Label Studio is an open-source data labeling tool designed for labeling, annotating, and exploring various data types. The tool also features a robust machine learning interface, which can be utilized for training new models, active learning, supervised learning, and various other training techniques.

Training a ResNet image classification model





Category	Annotations
Normal	3,962
Composite	1,486
Amalgam	660
Calculus	627
Caries	578
Root filling	239
Crown	139
Implant	7

How do you create labels?

Training data

- 995 multi-tooth images
- 8 categories
- 3,736 annotations

ResNet50 model

- Deep CNN
- Most popular architecture
- 20,000 + citations

Public data set
https://universe.roboflow.com/shauns-flow/dental-aq0l2

Labeling strategies

Getting labels for your model can be very time consuming

- Use annotated data sets, if available
- Automate the labeling process using EHRs

If you need manual labeling:

- Use a server approach (e.g., LabelStudio)
- Determine complexity of labels:

    Image classes

    Bounding boxes

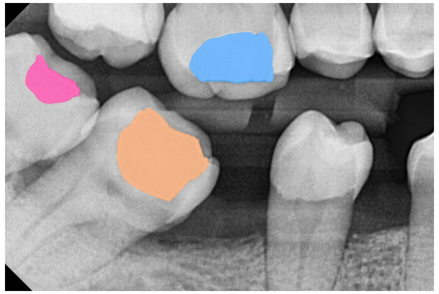
    Segmentation masks

- Combine multiple tasks in one label project

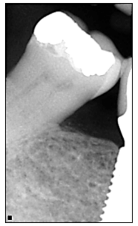
Image-level
“There is at least one tooth with an amalgam filling in this image”

Tooth-level
“Determine the type or restauration for each tooth”

Region-level
“Mark the outlines of the fillings in this image”





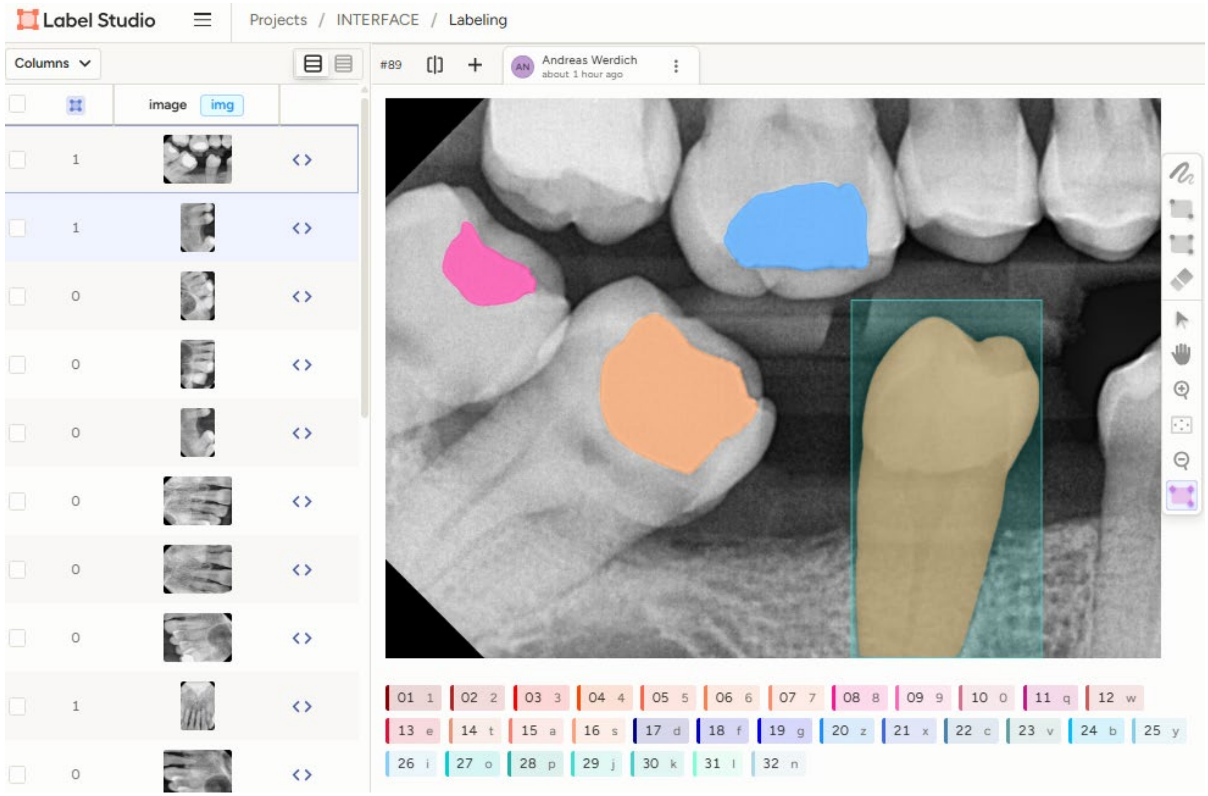



Label strategies for computer vision models

Manual labeling

Label Studio
Community Edition

- Open-source labeling tool
- Based on Python-Django
- Fully customizable label interface
- Can be run as a server in a secure network
- Access with VPN, 2FA
- Great for collaborative label efforts
- Simultaneous annotation by multiple users
- ML backend for automated pre-annotation



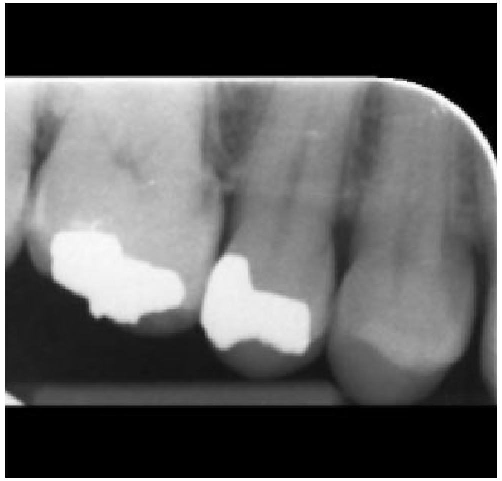
https://labelstud.io/

How good is the model

Best practices for testing

Given a new image that the model has not seen, how well does it predict the label?

- Before training: hold out a test set
- Do not use it to train or optimize the model
- Measure the performance



Label	Annotations
Control	30
Composite	30
Amalgam	30
Calculus	30
Caries	30
Root filling	30

Control	Composite	Amalgam	Calculus	Caries	Root filling	Ground truth
0.00455	0.131	$\boxed{0.627}$	0.134	0.101	0.00245	Amalgam

The model predicts Amalgam. That’s good.
How good is the model for detecting amalgam fillings in radiographs?

How good is the model?

Compare each class against the rest

- Converts strategy to binary classifications: “How well can the model predict the class label?”
- Get the probabilities for all test images with the ground truth label AMALGAM
- Set a threshold (e.g. 0.5)
- Count the number of correct and incorrect predictions
- Create the confusion matrix

Amalgam	Other classes
0.627	0.373

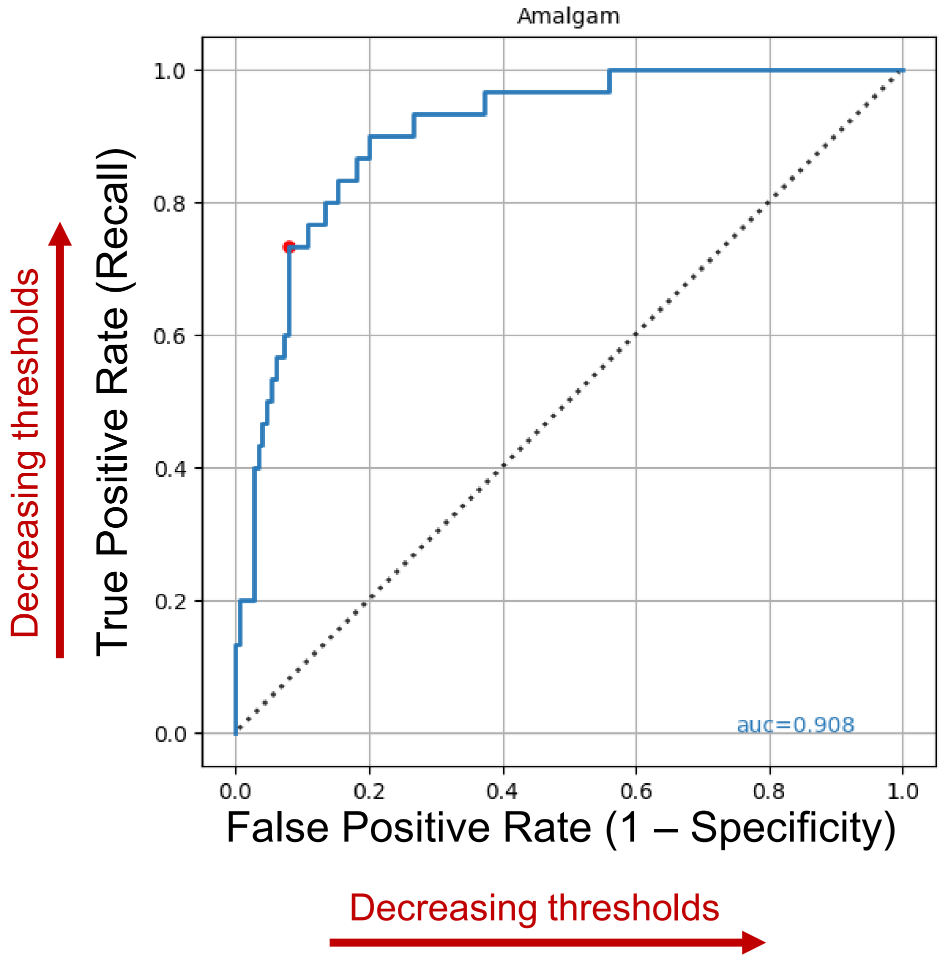
What is a good threshold?

		Predicted condition	
	Total population = P + N	Positive (PP) 34	Negative (PN) 146
Actual condition	Positive (P) 30	True positive (TP) 22	False negative (FN) 8
	Negative (N) 150	False positive (FP) 12	True negative (TN) 138

https://en.wikipedia.org/wiki/Confusion_matrix

Receiver – Operator Curve

Performance of the model for detecting the



How do you pick the classification threshold?

Low thresholds
When the cost of missed diagnosis is high (e.g., cancer)

High thresholds
When the cost of a false positive is high (e.g., invasive procedure)

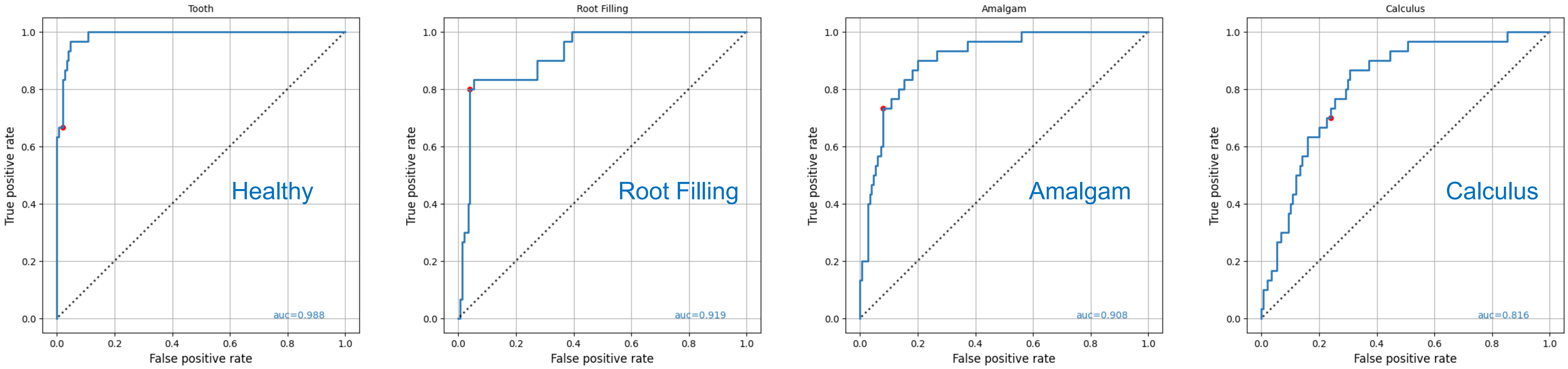
Strategy: Set the threshold for a fixed recall (e.g., 70%) and measure everything else at that threshold.

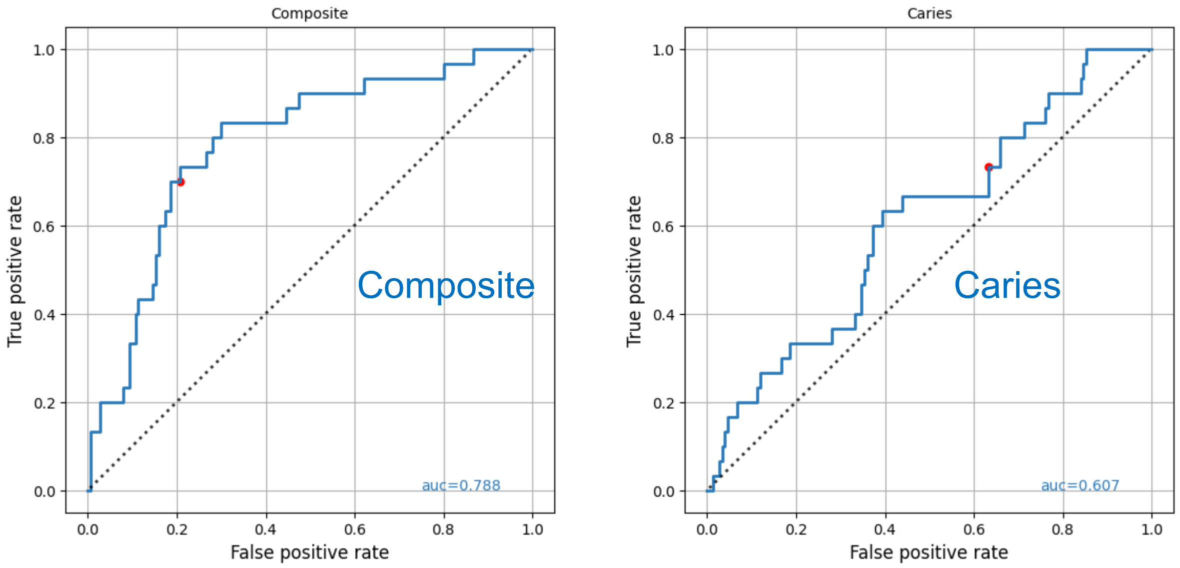
Most common performance metrics

- Accuracy: $\mathrm{ACC} = \frac{TP+TN}{P+N}$
- Recalll: $\mathrm{TPR} = \frac{TP}{P} = \frac{TP}{TP+FN}$ (sensitivity)
- Precision: $PPV = \frac{TP}{PP} = \frac{TP}{TP+FP}$
- Specificity: $TNR = \frac{TN}{N}$ (specificity)
- False positive rate: $FPR = \frac{FP}{N}$ (1 – specificity)
- AUC score (area under the ROC curve)
- F1 score: $F1 = 2\frac{Precision \times recall}{precision + recall}$ (harmonic mean)
- AUC score (area under the ROC curve)

https://en.wikipedia.org/wiki/Precision_and_recall

Results for the classification model





Model performance at 70% recall (sensitivity)

Label	AUC	PRECISION	F1 SCORE	TRAINING SAMPLES
Healthy	0.988	0.870	0.755	3,902
Root filling	0.919	0.800	0.800	177
Amalgam	0.908	0.647	0.688	600
Calculus	0.816	0.368	0.483	567
Composite	0.788	0.404	0.512	1,426
Caries	0.607	0.188	0.299	518

How can we understand the model decisions?

Explainable AI

Powerful method to visualize a model's decision

Idea: Visualize the features in the image that are most important for predicting the target class

"Where does the model see a person brushing teeth in the image?"

Gradient-weighted class activation mapping

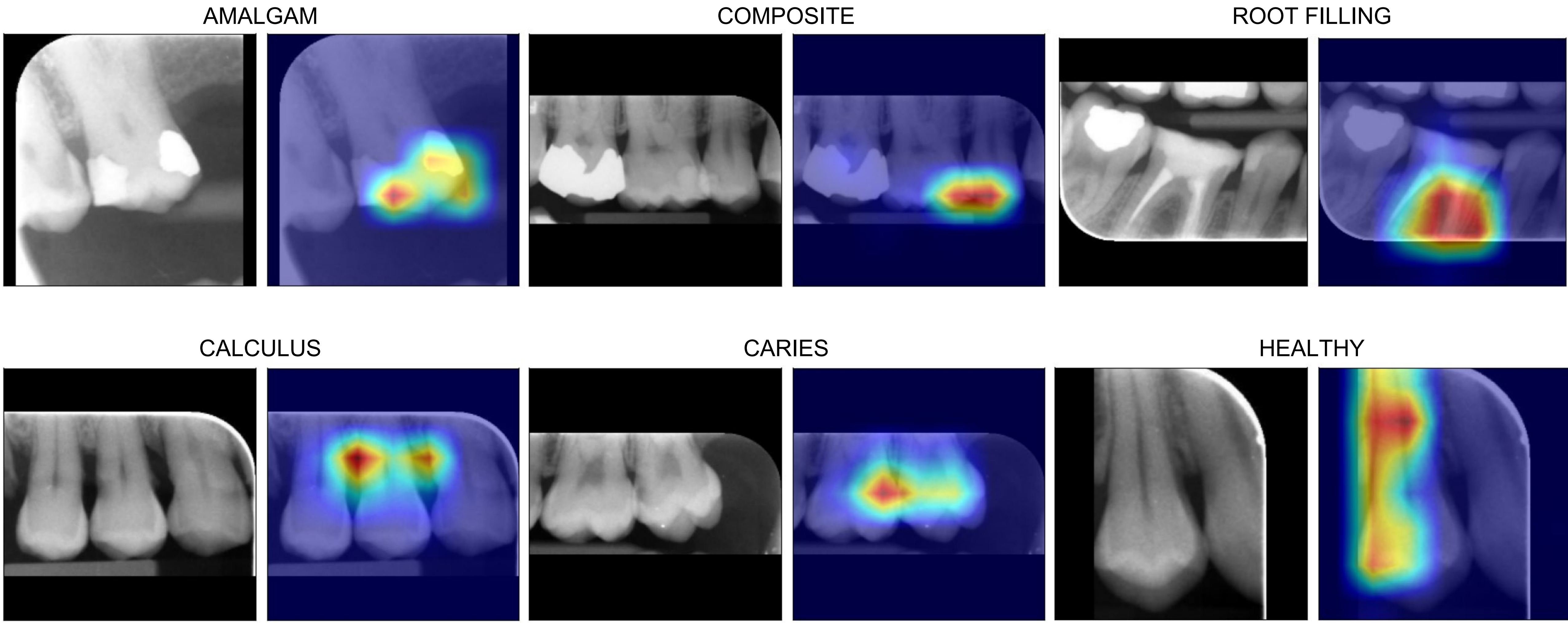
- Visualize the feature maps (filters) of the last conv layer
- Calculate the gradients of the target class in that layer

Process

- Run forward-pass on a test image
- Compute the gradients of the target class
- Form a weighted sum of the feature maps
- Create a heatmap
- Overlay the heatmap with the original image

Learning deep features for discriminative localization
Zhou et al. 2015 Computer Vision and Pattern Recognition:
https://arxiv.org/abs/1512.04150

Grad-CAM Examples



Activation Mapping

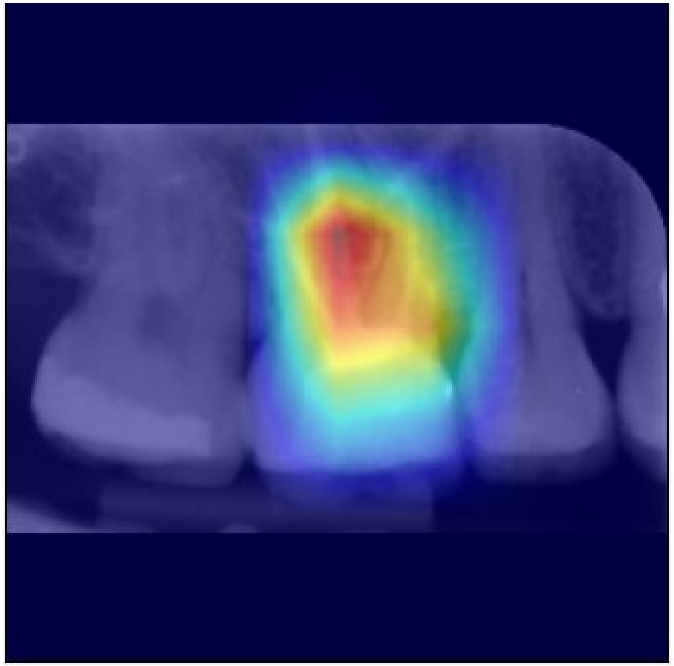
Helping to understand wrong predictions

TOOTH	AMALGAM	CARIES	COMPOSITE	CALCULUS	ROOT FILLING	CL	TRUE	PRED
0.178454	0.012511	0.158774	0.378917	0.053948	0.217397	5	ROOT FILLING	COMPOSITE

Common reasons for wrong predictions

- Performance (not enough training samples)
- At high performance
  - Artifacts (diagnosis labels on image)
  - Multiple classes in one image

Multiple labels in the image reduce the confidence score of the predicted class



Ground truth: Root Filling
Prediction: Composite

We need to train the model on single-tooth images!

Image segmentation

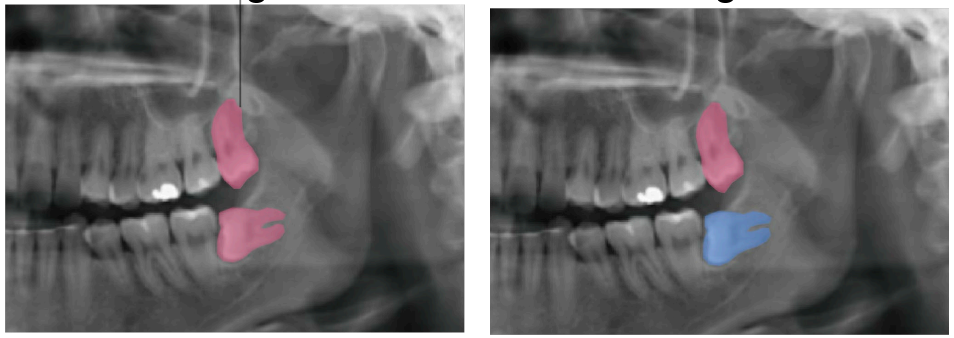
Extracting features from images

Semantic segmentation
- Classify each pixel in the image
- Not distinguishing between instances

Instance segmentation
- Assigns class and instance labels
- Identifies pixels belonging to different objects
- Perfect for identifying the borders of teeth

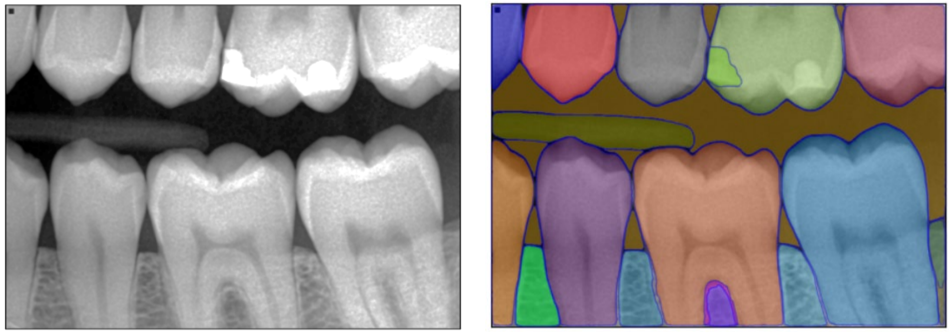
Panoptic segmentation
- Assigns both class and instance to every pixel
- Important for understanding the whole environment
- Requires rich annotations

Semantic segmentation Instance segmentation



(Adapted from **Feher, Tussie & Giannobile, 2024**)

SegmentAnything SAM



Instance segmentation with Detectron2

High-performance framework for training segmentation models

Detectron2 (2019)

- Instance segmentation
- Panoptic segmentation
- Object detection
- Keypoint detection

https://github.com/facebookresearch/detectron2

Installation

See installation instructions.

Getting Started

See Getting Started with Detectron2, and the Colab Notebook to learn about basic usage.

Learn more at our documentation. And see projects/ for some projects that are built on top of detectron2.

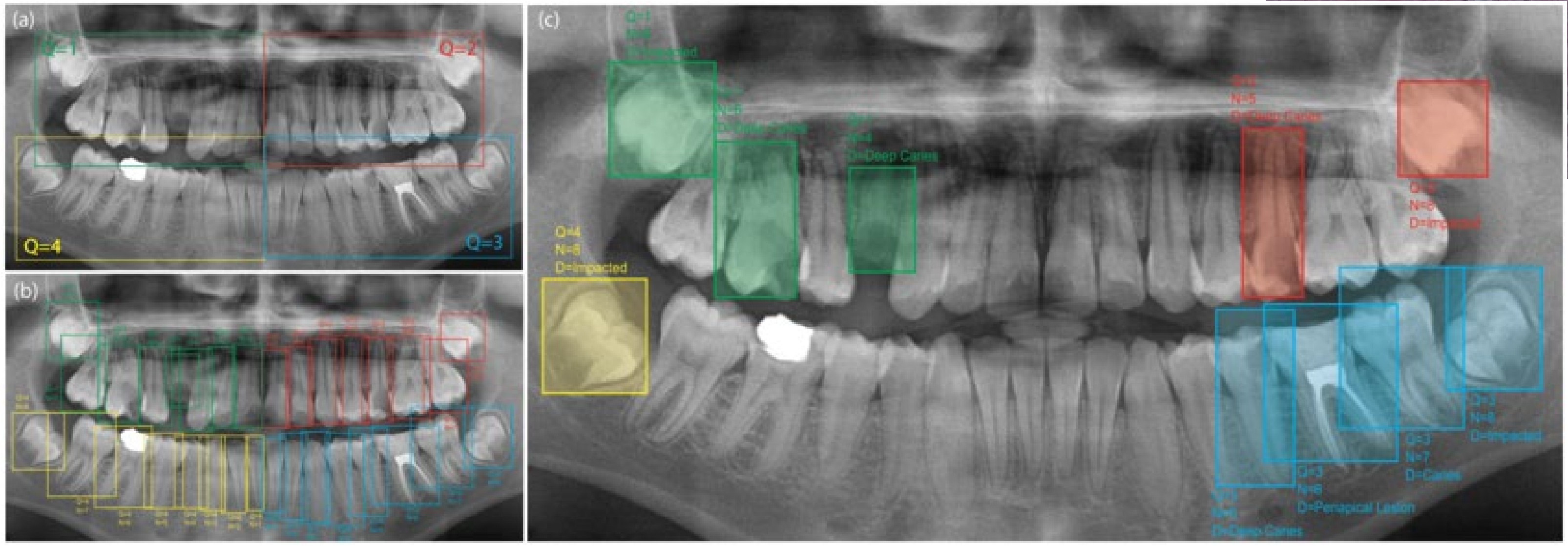
How do we get the segmentation labels?

1. If you have time: Make the effort to label images
2. If you don't: Use a public data set

Dentex: A public data set for tooth segmentation and detection

The dental enumeration and diagnosis challenge 2023

Goal: Encourage algorithms that can accurately detect abnormal teeth with dental enumeration and diagnosis
Public data set with 634 annotated panoramic radiographs,
> 20,000 annotations



Data split

- Training: 550 images
- Validation: 34 images
- Testing: 50 images

https://dentex.grand-challenge.org/
https://zenodo.org/records/7812323#.ZDQE1uxBwUG

Training the segmentation model

The Computer Vision Repository

The CCB Computer Vision Repository
https://github.com/ccb-hms/computervision/tree/main/notebooks/segmentation

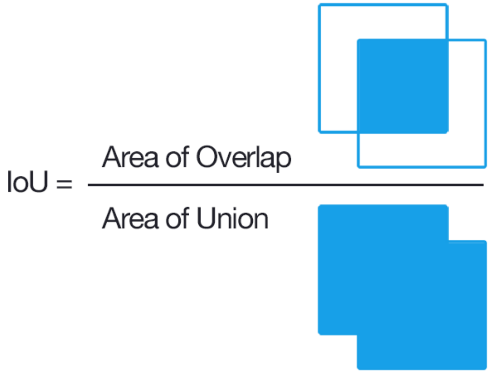
Includes 7 Jupyter notebooks for end-to-end model training:

- Download the panoramic data set
- Split into training, validation, and testing
- Format the annotations into Detectron2 format
- Train the tooth segmentation model
- Test the data set (calculate AP score)
- Perform segmentation on any input image

Details in part 2 of this lecture!

Result

Mask R-CNN model of the Detectron2 computer vision library



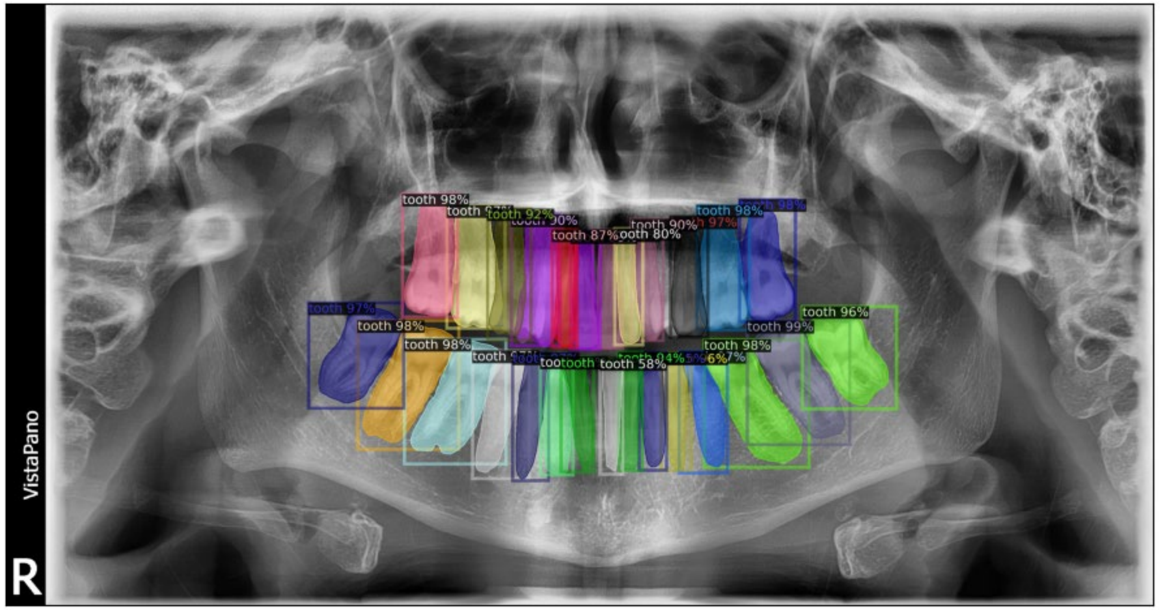
- Training with 550 panoramic radiographs
- Excellent performance on the test data
- See https://github.com/ccb-hms/computervision

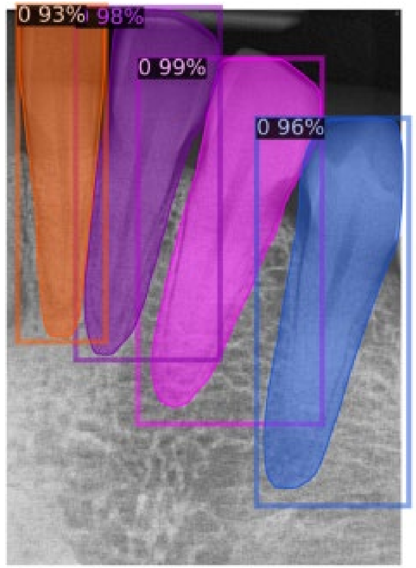
Model performance

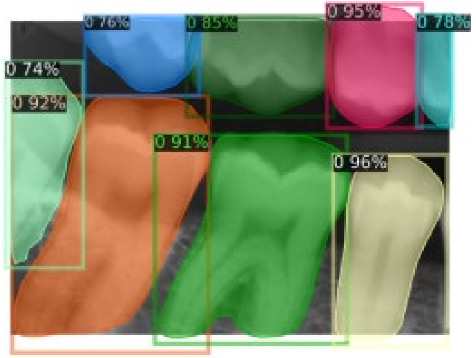
AP with IoU50: **0.9788**

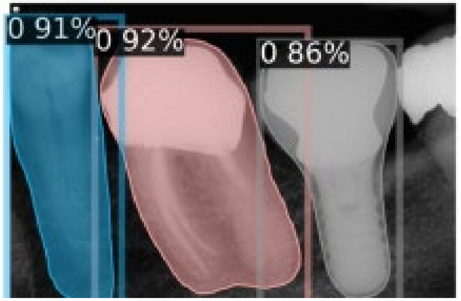
AP with IoU75: **0.5180**

In-house radiographs



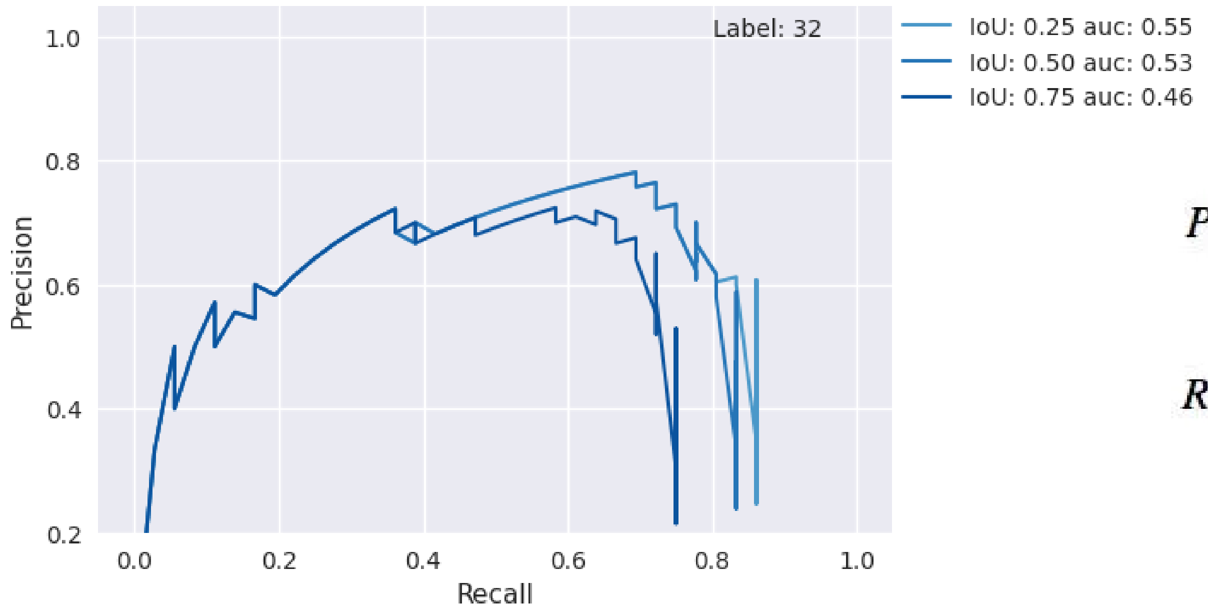






Average precision AP scores

Average Precision: area under the Precision – Recall curve



$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

Part II: Measure Mean Average Precision using a trained vision transformer model

Summary and outlook

- Convolutional Neural Networks: Most successful neural network architecture for computer vision
- Integral component of transformer-based architectures (vision transformers)
- Outperform multi-model LLMs
- Small, low-cost, trained on a single GPU using open-source software and data

Next lecture 07/15/2026

- How to get started with computer vision on your laptop
- The CCB computer vision repository: Template code for your computer vision projects
- Computer vision workflow
  - Coding environments, frameworks
  - Image pre-processing
  - Model training and evaluation
  - Inference

Thank you!

https://github.com/ccb-hms/computervision